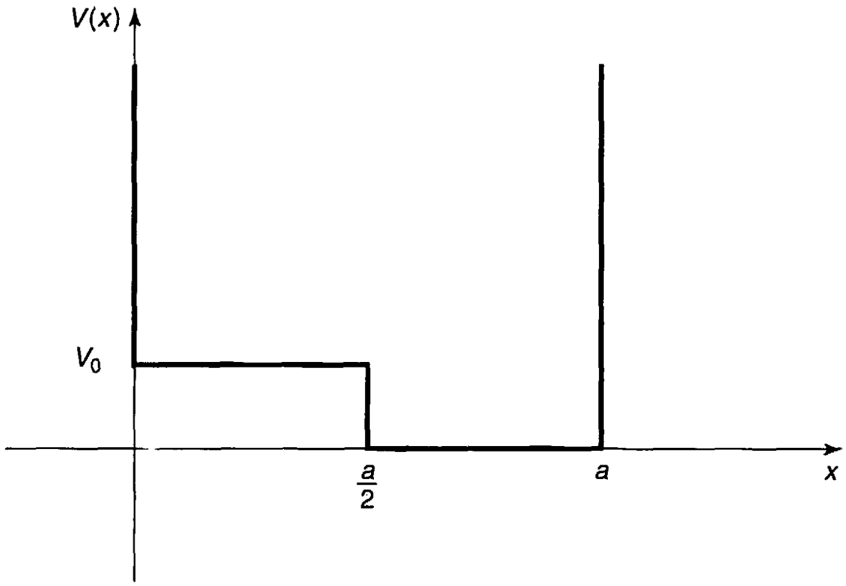
PROBLEM SET 4

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 70)

1) **(20 pts)** Use the WKB approximation to find the allowed energies E_n of an infinite square well with a "shelf", of height V_0 extending half-way across

$$V(x) = \begin{cases} V_0, & 0 < x < a/2 \\ 0, & a/2 < x < a \\ \infty, & \text{otherwise} \end{cases}$$



Express you answer in terms of V_0 and $E_0 \equiv (n\pi\hbar)^2/2ma^2$ (the n*th* allowed energy for the infinite square well with no shelf). Assume that $E_1^0 > V_0$, but do not assume that $E_n \gg V_0$.

2) **(20 pts)** Calculate the lifetimes (τ) of (a) **(10 pts)** U^{238} and (b) **(10 pts)** Po^{212}, using the derived equations in class, based on Gamow's theory of alpha decay,

$$\gamma \approx \frac{\sqrt{2mE}}{\hbar}\left[\frac{\pi}{2}r_2 - 2\sqrt{r_1 r_2}\right] = K_1\frac{Z}{\sqrt{E}} - K_2\sqrt{Zr_1}, \text{ for } r_1 \ll r_2 \text{ where}$$

$K_1 = 1.980\ \text{MeV}^{1/2}$, and $K_2 = 1.485\ \text{fm}^{-1/2}$, and the lifetime of the parent nucleus is $\tau = \dfrac{2r_1}{v}e^{2\gamma}$

Hint 1: The density of nuclear matter is relatively constant, so the nuclear radius empirically is $r_1 \approx (1.07\ \text{fm})A^{1/3}$, where A is the number of neutrons plus protons (in this example, it is for the parent nucleus).

Hint 2: The energy E of the emitted alpha particle can be obtained by using Einstein's formula, $E = m_p c^2 - m_d c^2 - m_\alpha c^2$, where m_p is the mass of the parent nucleus, m_d is the mass of the daughter nucleus, and m_α is the mass of the alpha particle (i.e. helium-4 nucleus). To figure out the mass of the daughter nucleus, note that since the alpha particle carries off two protons and two neutrons, the parent nucleus number of protons Z decreases by 2, and its mass number A decreases by 4. (Look up the relevant nuclear masses online)

Hint 3: To estimate the alpha particle speed (v), use $E = (1/2)m_\alpha v^2$, which ignored the potential energy inside the nucleus and underestimates v, but it is the best approximation we can make.

3) **(30 pts)** A particle of fixed energy $E < V(x)$ moves within a harmonic oscillator with potential $V(x) = \frac{1}{2}m\omega^2 x^2$

(a) **(5 pts)** Make a qualitative plot of $V(x)$ versus x and E overlayed

(b) **(5 pts)** Determine the turning points (x_1, x_2), where the "classical" region joins the "nonclassical" region where the WKB approximation breaks down

(c) **(20 pts)** Use the WKB quantization condition for the case of a potential well with two sloping sides derived on lecture (using connection formulas)

$$\int_{x_1}^{x_2} p(x)dx = \left(n - \frac{1}{2}\right)\pi\hbar, \text{ where } p(x) = \sqrt{2m[E - V(x)]}$$

and determine the allowed energies E_n of the harmonic oscillator